

Origin Pilot: The First Quantum Operating System**Introduction**

The field of Quantum computing is vast ever-changing industry. It is revolutionizing the problems that were thought to be impossible for any computer, into tasks that seem more like warm up tasks. From this industry and sub field of quantum operating systems are emerging. The Chinese company Origin Quantum had first launched in operating system in 2021 and calls it Origin Pilot. This operating system has been released in February 2026. Origin Pilot had become China's first ever quantum operating systems and is now acting as the central command hub for quantum computing operations and scheduling. Origin Pilot is compatible with the following hardware type: Superconducting, Trapped ions, Semiconductors, Neutral atoms, and Photonicsⁱ.

The main workflow of Origin Pilot works as follows. The compiler can identify the quantum part and classic part with lexical, grammar and semantic analysis. The quantum program compiled to a quantum application. If the hybrid program is written in distributed computing programming languages, the Origin Pilot will send the classic computing tasks to high performance computing clusters with a classic job scheduling system. The quantum computing part will be sent to the quantum task scheduling service. These programs all become hybrid with the help of QRunes. A diagram of the overall workflow can be referenced in the first item under in the [appendix](#).

The biggest hurdle for a quantum OS is the ability to integrate scheduling, orchestration of hardware and software, multi-threading and automatic qubit shifting. These are the just the main tasks that Origin Quantum has been able to achieve with the conception of Origin Pilot. Quantum insider states that, "Chinese researchers describe the release as both a technical milestone and a strategic step in the country's effort to strengthen self-reliance in critical technologies,"ⁱⁱ. These efforts are not to go unnoticed some of the biggest accomplishments are of task scheduling, parallel exaction, and automatic calibration.

Task Scheduling

There are always numerous tasks that need to be execute on any computer for it to function effectively. For classical computers, there are different scheduling algorithms are used to manage jobs to run from some type of queue. The algorithm is designed in a specific way that determines the ordering of executing jobs.

This was not something accomplished in quantum until Origin Quantum. Like task on a classical computer, Origin Pilot has 5 characteristics for each task that is ready for execution. These include, the number of qubits required, the programs' immediate representation, the ID of the processor, the type of task either general or some sort of calibration, and the priority of the quantum task, itselfⁱⁱⁱ.

There are a few different algorithms that can be using within a classical computer to handle task scheduling. For Origin Pilot the algorithms that use is referred to as Highest Response Ratio Next, or HRRN, This algorithm considered both the run time and wait time in each task and the priority is determined by the following equation:

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$$R_p = \frac{T_{waiting_time} + T_{runtime}}{T_{runtime}} = \frac{T_{response_time}}{T_{runtime}}$$

Parallel Execution

In the current state of Origin Pilot, synchronous parallelisation is applied to preform parallel execution, this combines multiple different tasks into one big quantum transaction. Quantum tasks are scheduled and executed in parallel using transactions, but only one transaction runs at a time. A diagram of load balancing can be references within the second items in the [appendix](#). One of the biggest mitigations preformed has to do with entanglement. Swap gates are introduced and theses gates help to route qubits into positions where entanglement operations are possible. These gates are not directly mitigating the entanglement limitation, but indirectly preserve it at the cost of increase error due to the additional SWAP gate decompositions.

Based on the parallel execution there are a few various scheduling differences due to parallel execution. First, automatic calibration – discussed further in the next section- are given higher priorities and do not require qubit mapping which helps to prevent interference with other tasks. Also, calibration tasks follow a First-Come-First-Served (FCFS) scheduling strategy. The mapping process involves converting the quantum program into a Directed Acyclic Graph (DAG) and identifying optimal subgraph sequences.

Automatic Calibration

The two main parameters of a single qubit are its coherence time and gate fidelity. The coherence times can help determine the coupling strength between the system of the qubit and the quantum environment that it is in. The quality of the qubit is dependent on several different factors; therefore continual calibration of the qubits is required.

Within the process of calibration consists of two different steps. One of them is the performance checking of the qubits. The other step is the process of adjustment of errors. Quantum Ogrin has built a partial observable Markov Decision Process to perform the automatic qubit calibrations needed. These kinds of tasks are split up in specific calibration and execution regions; this is all handles by the processor itself. This helps to ensure that tasks run on the best selected region and in doing so improves the fidelity of the qubit, too.

Conclusion

Even with the release of Ogrin Pilot as an open-source software, is still in the early stages of its development. Some of the biggest achievements that Quantum Origin has accomplished with this OS task scheduling, parallel execution, and calibration, with another one being , resource management but not referenced in this report. Some of the future work includes distributed quantum computing and hybrid classical-quantum systems. The plan back in 2021 was for an open-source public release of the software, which was recently accomplished in February 2026, only about 2 months ago.

Origin Pilot: The First Quantum Operating System**Appendix**

[1] From: "Origin pilot: a quantum operating system for efficient usage of quantum resources."

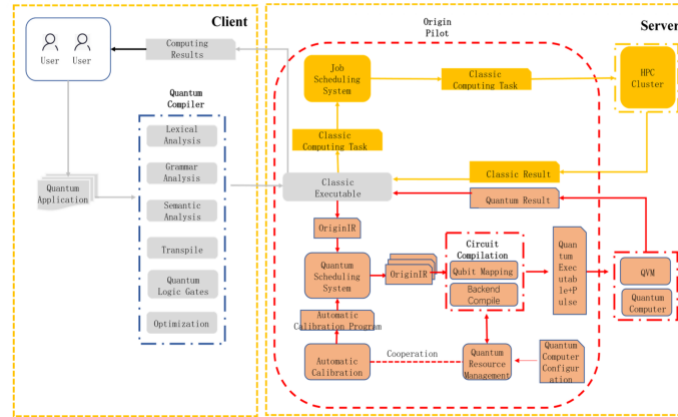


FIG. 2. Workflow of Origin Pilot

[2] From: "Origin pilot: a quantum operating system for efficient usage of quantum resources."

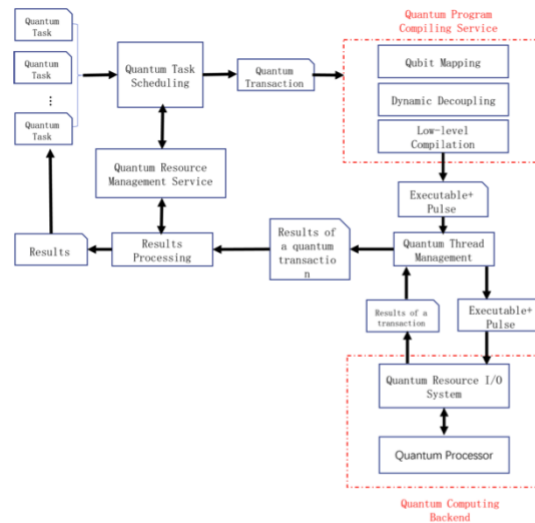


FIG. 4. Multiple quantum processors' load balancing

ⁱ "Origin Pilot: a Quantum Operating System for Efficient Usage of Quantum Resources," Origin Quantum, accessed April 23, 2026, <https://originqc.com/blogs/origin-pilot-download>.

ⁱⁱ "Chinese Firm Releases Open-Source Quantum Operating System for Public Download," The Quantum Insider, February 27, 2026, <https://thequantuminsider.com/2026/02/27/chinese-firm-releases-open-source-quantum-operating-system-for-public-download/>.

ⁱⁱⁱ Kong, Weicheng, Junchao Wang, Yongjian Han, Yuchun Wu, Yu Zhang, Menghan Dou, Yuan Fang, and Guoping Guo. "Origin pilot: a quantum operating system for effecient usage of quantum resources." arXiv preprint arXiv:2105.10730 (2021), pg 6.